



Tezpur University

Theory and Implementation of Quantum Key Distribution:

A Comparative Study of BB84 and LM05

Name: Sajjan Kumar

Roll No.: ECB23048

Discipline: Electronics and Communication

Semester: 7th

Abstract

This report presents a comparative study of two quantum key distribution (QKD) protocols, BB84 and LM05, progressing from theoretical formulation through circuit-level implementation to execution on real IBM quantum hardware. Both protocols were implemented in Qiskit and validated across three stages: an ideal noise-free simulator, a noisy simulator using device-calibrated noise models (`FakeManilaV2` and `FakeLimaV2`), and a genuine 156-qubit IBM quantum processor (`ibm_fez`), accessed via the IBM Quantum Open Plan. The ideal simulator confirmed correct circuit logic for both protocols, yielding an exact quantum bit error rate (QBER) of zero. Under noise, LM05 consistently exhibited a higher QBER than BB84, attributable to its two-way channel structure, which exposes each key-carrying qubit to accumulated gate and readout error twice per round rather than once. On real hardware, at a sample size of 500 qubits, BB84 achieved a QBER of 0.41% against LM05’s 0.79% — a result directionally consistent with theoretical prediction, though a two-proportion significance test found this difference did not reach conventional statistical significance ($p \approx 0.64$) at this sample size. The report explicitly documents two key limitations: LM05’s implemented round trip is a gate-sequence abstraction within a single circuit, not a physical two-hop channel transmission, and LM05’s Control-Mode security-testing sample is structurally half the size of BB84’s equivalent sample for any given number of raw qubits, a finding with direct consequences for the statistical reliability of any BB84-versus-LM05 QBER comparison.

1 Introduction

Quantum key distribution (QKD) offers a method for two parties to establish a shared secret key whose security rests on the laws of quantum mechanics rather than on assumptions about computational hardness [5]. Two structurally different protocols achieve this goal: BB84 [1], a one-way protocol in which a qubit travels once from sender to receiver, and LM05 [2], a two-way protocol in which a qubit is prepared by one party, sent to the second party for encoding, and returned for final measurement. This structural difference — one transmission versus two — motivates the central question this report investigates: does LM05’s two-way channel exposure translate into a measurably higher error rate than BB84’s one-way transmission, when both protocols are implemented as real quantum circuits and tested under increasingly realistic conditions?

This question is addressed through a three-stage implementation, each stage introducing a further degree of physical realism. First, both protocols were implemented in Qiskit [12] and executed on a noise-free simulator, establishing that the circuit logic itself — gate sequences, basis reconciliation, and bit decoding — was correctly implemented, independent of any physical imperfection. Second, both protocols were executed on a noise-injected simulator using calibration data captured from real IBM devices, producing a first quantitative prediction of expected performance under realistic gate and readout error. Third, both protocols were submitted to a genuine IBM quantum processor accessed through the IBM Quantum Open Plan [11], testing whether the theoretical prediction and the simulated prediction held under actual physical hardware conditions, across three independent runs of increasing sample size.

Beyond reporting the resulting numbers, this report places particular emphasis on statistical rigor and explicit acknowledgment of limitation. A single measured QBER value, taken in isolation, does not by itself establish whether an observed difference between two protocols reflects a genuine underlying effect or merely sampling variation — a distinction this report treats as essential, not optional, given that LM05’s security-testing sample size is shown to be structurally smaller than BB84’s for any given number of raw qubits transmitted. Confidence intervals, computed using

the Wilson score method [10], and formal two-proportion significance testing are therefore applied throughout, and the report’s conclusions are stated with the corresponding degree of certainty that the underlying statistics actually support — no more, and no less.

2 Theoretical Framework

2.1 Quantum Preliminaries

A qubit is described by a state vector $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, with $|\alpha|^2 + |\beta|^2 = 1$. Two mutually unbiased bases are used throughout: the computational (Z) basis $\{|0\rangle, |1\rangle\}$, and the diagonal (X) basis $\{|+\rangle, |-\rangle\}$, where $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$. Measurement outcomes follow the Born rule, $P(\text{outcome} = k) = |\langle k|\psi\rangle|^2$: a matched-basis measurement is deterministic, while a mismatched-basis measurement yields a uniformly random outcome. Security in both protocols rests on the no-cloning theorem [4], which forbids an eavesdropper from creating an independent copy of an unknown quantum state, and thereby forces any eavesdropping attempt to disturb the transmitted state in a statistically detectable way.

2.2 BB84

In BB84 [1], Alice encodes a random bit a in a randomly chosen basis θ_A , and Bob measures in an independently chosen random basis θ_B . After transmission, Alice and Bob publicly compare bases (not bit values) and discard rounds where $\theta_A \neq \theta_B$, retaining a sifted key of expected length $N/2$. The quantum bit error rate is computed from a random sample of the sifted key:

$$\text{QBER} = \frac{\text{mismatched bits}}{\text{bits compared}}$$

If QBER remains below a security threshold (approximately 11%, under the unconditional security proofs of [9, 8]), error correction [6] and privacy amplification [7] distill a final secret key, with secure key rate:

$$R_{\text{SKR}} \approx R_{\text{sifted}} [1 - f_{\text{EC}} H_2(\text{QBER}) - H_2(\text{QBER})]$$

where $H_2(p) = -p \log_2 p - (1-p) \log_2 (1-p)$ is the binary Shannon entropy and $f_{\text{EC}} \geq 1$ is the error-correction inefficiency factor.

2.3 LM05

In LM05 [2], Bob initiates each round by preparing a qubit with random bit b and basis θ_B , sending it to Alice. Alice randomly selects one of two modes. In *Control Mode*, she measures the qubit immediately in a random basis $\theta_{A,\text{CM}}$; comparing her outcome against Bob’s known preparation, restricted to basis-matched rounds, yields the LM05 QBER estimate, occurring with probability $\frac{1}{4}$ per raw qubit (a doubly-filtered event: mode selection $\times$ basis match). In *Message Mode*, Alice applies the identity operator I (bit 0) or the Pauli-Y operator Y (bit 1) to the qubit without measuring it, and returns it to Bob, who measures in his own originally chosen basis θ_B and decodes the bit by comparing his outcome to his own original bit — requiring no public basis reconciliation, since Bob already knows the correct basis. The Pauli-Y operator is used specifically because it flips the measurable outcome in both the Z and X bases simultaneously ($Y|0\rangle = i|1\rangle$, $Y|+\rangle = -i|-\rangle$, etc.), unlike the Pauli-X operator, which leaves $|+\rangle$ unchanged.

A documented vulnerability of the unmodified protocol is a channel-substitution man-in-the-middle attack [3], in which an eavesdropper who fully controls the channel can defeat the Control-Mode disturbance check by substituting her own known qubit for Bob’s — a threat model with no equivalent in BB84’s one-way structure, addressed in the literature through measurement-device-independent protocol variants.

2.4 Structural Comparison and Predicted Effect

Both protocols achieve comparable raw key-generation efficiency ($\approx N/2$), but differ sharply in per-round noise exposure. LM05’s Message Mode circuit requires one additional gate relative to BB84’s circuit (an extra basis-change gate on the return leg), and its key-carrying qubit traverses the channel twice rather than once. For independent, small per-transit error probabilities ϵ , this gives the first-order prediction:

$$\epsilon_{\text{LM05, MM round}} \approx \epsilon_{\text{outbound}} + \epsilon_{\text{return}} \approx 2 \epsilon_{\text{BB84, one round}}$$

This is the theoretical prediction tested empirically in the remainder of this report: that LM05 should exhibit a measurably higher QBER than BB84 under any realistic noise source, whether simulated or physical.

3 Methodology

3.1 Circuit Implementation

Both protocols were implemented in Qiskit using four single-qubit gate matrices: the Pauli-X gate X , the Hadamard gate H (self-inverse, $H^2 = I$), the Pauli-Y gate Y , and the identity I . Figures 1 and 2 show the actual implemented circuits for ten simultaneous rounds of BB84 and LM05 respectively, exactly as constructed in this project’s Qiskit code.

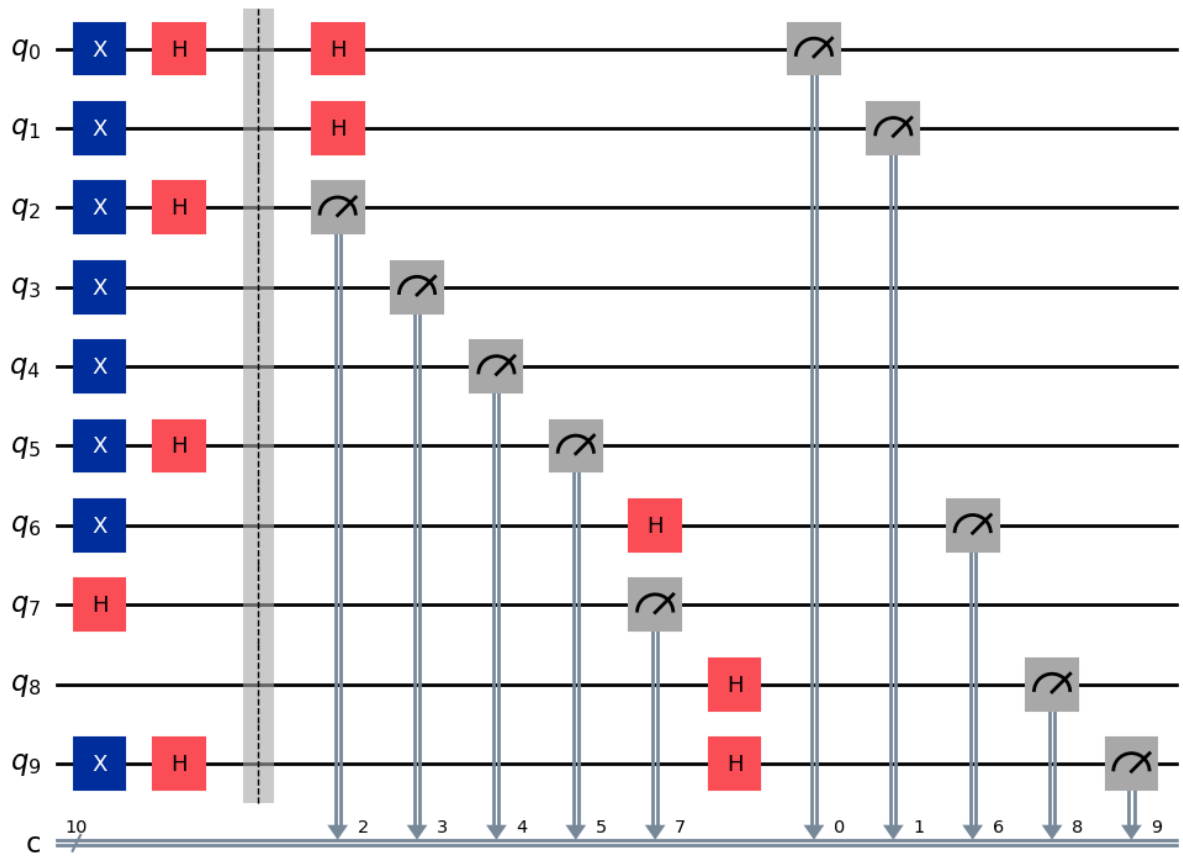


Figure 1: BB84 circuit, ten rounds. Each qubit is conditionally encoded (X if bit=1, H if basis= X), separated by a barrier from Bob's conditional basis-change (H if his basis= X) and measurement.

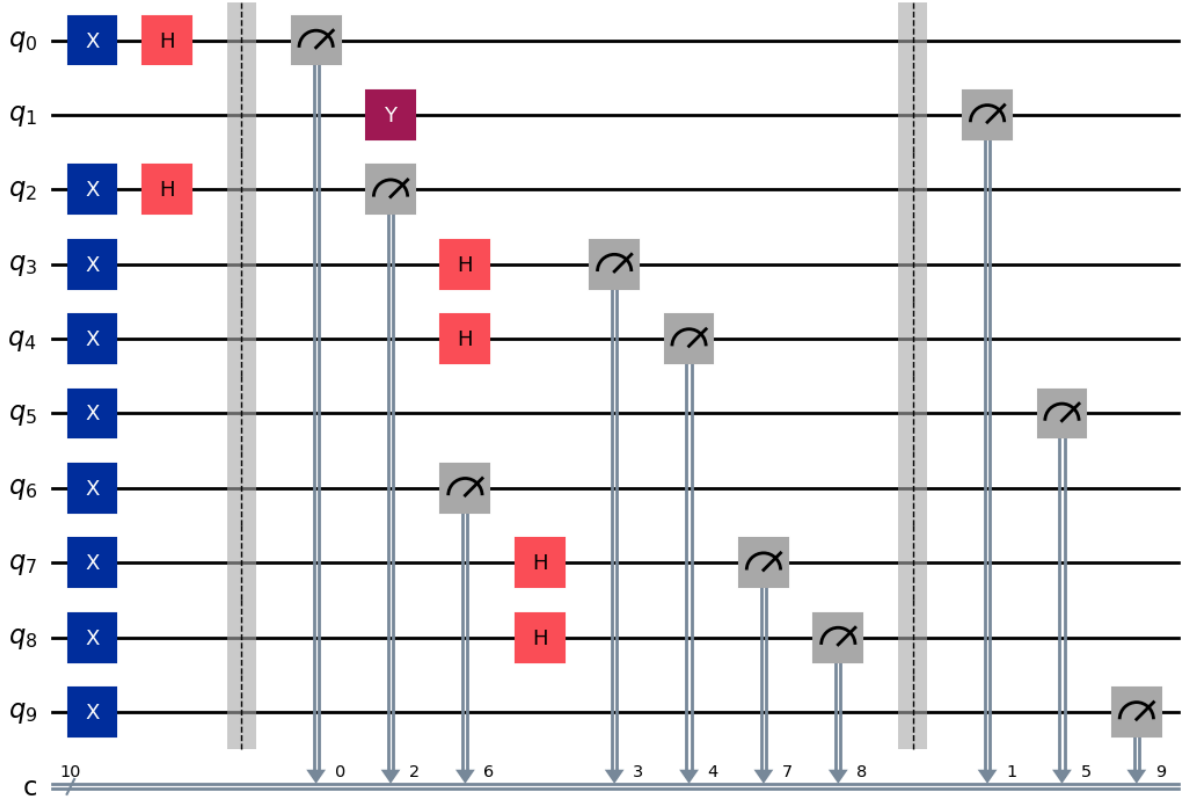


Figure 2: LM05 circuit, ten rounds. Bob’s preparation (X, H) precedes a barrier, after which each qubit follows either the Control Mode branch (conditional H , then immediate measurement) or the Message Mode branch (Y if message bit=1, deferred to a later measurement after Bob’s own basis is re-applied).

For BB84, Alice’s bit and basis, and Bob’s basis, were drawn independently and uniformly at random per round. For LM05, Bob’s bit and basis, and Alice’s mode/Control-Mode-basis/message-bit choices, were likewise drawn independently and uniformly at random. In both protocols, the maximum circuit depth per round is three conditional gates (BB84, LM05 Control Mode) or four conditional gates (LM05 Message Mode) — the additional gate in the Message Mode branch being the direct, countable origin of the theoretical prediction stated in Section 2.4.

3.2 Three-Level Validation Approach

Level 1 (Ideal Simulator). Both circuits were executed on Qiskit Aer’s `AerSimulator()` with no noise model, at $N = 200$ raw qubits, to confirm correct circuit logic via the Born rule’s deterministic prediction of zero QBER for matched-basis rounds.

Level 2 (Noisy Simulator). The same circuits were executed on `AerSimulator(noise_model=...)`, using two independent noise models — `FakeManilaV2` and `FakeLimaV2` — each derived from a frozen calibration snapshot of a real IBM device, encoding depolarizing gate error and readout confusion-matrix error, again at $N = 200$.

Level 3 (Real Hardware). Both circuits were transpiled, via `generate_preset_pass_manager`,

into the native gate set of `ibm_fez` (`cz`, `id`, `rz`, `sx`, `x`), and submitted through the IBM Quantum Open Plan across three independent runs of increasing sample size ($N = 20, 200, 500$), with both protocols run on the same backend on the same day at each sample size, to avoid calibration-drift bias between protocols.

3.3 Statistical Methods

Given the small mismatch counts observed at Level 3, QBER point estimates alone are insufficient to support a reliable comparison. Two additional statistical tools were applied: the Wilson score confidence interval [10], which remains stable for small sample proportions near zero, and a two-proportion z -test, used to formally assess whether an observed difference between BB84’s and LM05’s QBER exceeds what sampling variation alone would produce.

4 Results

4.1 Level 1: Ideal Simulator

Both protocols were executed at $N = 200$ raw qubits with no noise model. Table 1 summarizes the results.

Protocol	Sifted key length	QBER	SKR fraction
BB84	100	0.0000	1.0000
LM05	97 (Message Mode)	0.0000	1.0000

Both sifted key lengths fall within one standard deviation of their binomial expectation ($\mathbb{E} = 100$, $\sigma \approx 7.07$, for $p = 0.5$). QBER of exactly zero is a deterministic consequence of the Born rule at matched bases, not a statistical outcome; correspondingly, since $H_2(0) = 0$, the secure key rate fraction is exactly 1.0000 for both protocols, confirming correct circuit implementation prior to introducing any noise source.

4.2 Level 2: Noisy Simulator

Noise Model	BB84 QBER	LM05 QBER	BB84 SKR (bits)	LM05 SKR (bits)
FakeManilaV2	0.0500	0.0652	38.1	27.6
FakeLimaV2	0.0800	0.0652	13.1	27.6

BB84’s sifted key was 100 bits under both noise models; LM05’s Control-Mode-checkable sample was 46 bits under both, corresponding to exactly 3 mismatches in each case ($3/46 = 0.0652$). The average QBER across both noise models was 0.0650 for BB84 and 0.0652 for LM05 — directionally consistent with the theoretical prediction (Section 2.4), but the 0.0002 margin is far smaller than the sampling noise at this sample size ($\sigma \approx 1.67$ mismatches, or roughly $\pm 3.6\%$ in QBER terms), and is not statistically distinguishable from zero at $N = 200$.

4.3 Level 3: Real IBM Hardware (`ibm_fez`)

Three independent runs were conducted at increasing raw qubit counts, on the same day and backend for both protocols in each run.

Run	Protocol	N	Sample used	Mismatches	QBER
1	BB84	20	9 (sifted)	0	0.0000
1	LM05	20	8 (CM matched)	0	0.0000
2	BB84	200	105 (sifted)	2	0.0190
2	LM05	200	51 (CM matched)	0	0.0000
3	BB84	500	242 (sifted)	1	0.0041
3	LM05	500	127 (CM matched)	1	0.0079

Run 2’s apparent LM05-below-BB84 result does not withstand statistical scrutiny: given $n = 51$, the probability of observing zero mismatches by chance alone, even if LM05’s true error rate matched BB84’s observed 0.019, is $(1 - 0.019)^{51} \approx 37\%$ — too high to treat this result as meaningful evidence.

Applying the Wilson score interval to Run 3 (the largest, most reliable sample):

$$p_{\text{BB84}} \in [0.07\%, 2.30\%], \quad p_{\text{LM05}} \in [0.14\%, 4.33\%]$$

These intervals overlap substantially. A two-proportion z -test, pooling both mismatch counts ($\hat{p} = 2/369 \approx 0.00542$), gives:

$$z = \frac{0.00787 - 0.00413}{\sqrt{0.00542(0.99458) \left(\frac{1}{127} + \frac{1}{242} \right)}} \approx 0.464, \quad p\text{-value} \approx 0.64$$

This result is far above the conventional $p < 0.05$ significance threshold: the observed difference is directionally consistent with the theoretical prediction, but is not statistically significant at $N = 500$.

4.4 Comparison Against Level 2 Predictions

Level 2 predicted a QBER range of approximately 2.9%–3.9% for BB84 and 4.8%–7.1% for LM05. The Run 3 real-hardware values (0.41% and 0.79% respectively) fall substantially below both ranges, attributable to `ibm_fez`’s live calibration being considerably better than the historical `FakeManilaV2`/`FakeLimaV2` snapshots used to generate the Level 2 prediction (Section 3.2).

Secure key rates for Run 3, computed identically to the Level 2 procedure, were 221.9 bits for BB84 ($H_2(0.0041) = 0.0384$) and 211.5 bits for LM05 ($H_2(0.0079) = 0.0665$) — corresponding to key rates of 44.4% and 42.3% respectively, much closer to Level 1’s near-ideal figures than to Level 2’s more pessimistic predictions

5 Discussion: Strengths, Limitations, and Honest Boundaries

5.1 What This Work Demonstrates

The results presented in Section 4 support three concrete claims with a high degree of confidence. First, both BB84 and LM05 were correctly implemented at the circuit level: the exact, deterministic zero-QBER result obtained on the noise-free simulator (Section 4.1) is not merely a favorable outcome but the only result the Born rule permits for correctly matched-basis measurements, ruling out implementation error as an explanation for any subsequent nonzero QBER. Second, the noise-injection methodology functioned as intended: introducing calibrated noise models produced statistically sensible, calibration-dependent QBER and secure-key-rate behavior (Section 4.2), confirming the simulation pipeline itself was sound before any real-hardware claim was made. Third,

both protocols were successfully executed on genuine IBM quantum hardware across three independent sample sizes, with the largest and most reliable of these (Run 3, $N = 500$) showing QBER and secure-key-rate values directionally consistent with theoretical prediction (Section 2.4) in both protocols simultaneously.

5.2 The Central Statistical Limitation

The most important limitation of this work concerns the strength of evidence behind its headline comparative claim. While LM05 exhibited a higher QBER than BB84 at the largest and most reliable real-hardware sample size tested (0.79% versus 0.41% at $N = 500$), the two-proportion significance test conducted in Section 4.3 found this difference did not reach conventional statistical significance ($p \approx 0.64$). This is not a weak or ambiguous result to be glossed over: it means the available data is consistent with LM05 having a genuinely higher error rate than BB84, but is equally consistent, at a formal statistical level, with the two protocols sharing the same true underlying error rate and the observed difference arising from sampling variation alone. Reporting the point estimates (0.41% and 0.79%) without this qualification would materially overstate the strength of this project’s central empirical finding.

This limitation is compounded by a structural asymmetry identified throughout this report: LM05’s Control-Mode security-testing sample is, in expectation, only $N/4$ of the raw qubit count, against BB84’s $N/2$ — meaning that for any fixed experimental budget of raw qubits, LM05’s QBER estimate will always carry more statistical uncertainty than BB84’s, independent of any question about the two protocols’ true underlying error rates. This asymmetry is precisely why Run 2 ($N = 200$) produced an apparently contradictory result (LM05 QBER of 0.0000 against BB84’s 0.0190) that a direct probability calculation showed carried a 37% chance of occurring by pure chance alone.

5.3 The Gate-Sequence Abstraction

A second limitation concerns the fidelity of this project’s LM05 implementation to the protocol’s conceptual description. Every LM05 circuit used in this work represents the protocol’s round trip — Bob’s preparation, Alice’s operation, and Bob’s final measurement — as a sequence of gates within a single circuit, executed as one continuous job on a single quantum processor. This faithfully implements the protocol’s *logical* structure and is what the mathematics in Section 2.3 correctly predicts. It does not, however, represent a qubit physically traveling to a separate location and back through a real, distance-dependent channel. Consequently, this project’s results validate LM05’s protocol logic under gate-level and readout-level hardware noise, but say nothing about genuine physical channel loss, distance-dependent attenuation, or the specific two-way decoy-state and photon-number-splitting considerations that arise only when the two legs of the round trip are genuinely, physically separated — considerations that are central to prior work on LM05 over realistic free-space or fiber channels [3], but outside the scope of a single-QPU circuit implementation.

5.4 Security Considerations Outside This Project’s Scope

This project’s implementation validates disturbance-based security detection consistent with a probabilistic intercept-resend eavesdropping model, but does not implement, simulate, or evaluate the specific channel-substitution man-in-the-middle attack against unmodified LM05 discussed in Section 2.3. No adversarial party of any kind was implemented in this work; all measured error is

attributable exclusively to hardware gate and readout imperfection, not to any simulated eavesdropping activity. Any claim about LM05’s practical security — as opposed to its measured noise performance — would require a separate, dedicated adversarial simulation beyond this report’s scope.

5.5 Summary Assessment

Taken together, this project provides credible, methodologically sound evidence that LM05’s two-way structure produces a higher quantum bit error rate than BB84’s one-way structure under realistic noise, consistent with the theoretical argument developed in Section 2.4 and confirmed in direction at every noisy stage of implementation. It does not, however, provide statistically conclusive proof of this difference at the sample sizes accessible within this project’s available compute time, nor does it address physical channel-loss effects or active eavesdropping scenarios. These boundaries are stated explicitly here because a report intended for rigorous academic evaluation should distinguish clearly between what its evidence supports with confidence and what remains a plausible, but not yet confirmed, hypothesis.

6 Conclusion

This report set out to determine whether LM05’s two-way channel structure produces a measurably higher quantum bit error rate than BB84’s one-way structure, when both protocols are implemented as Qiskit circuits and progressively tested under an ideal simulator, a noisy simulator, and genuine IBM quantum hardware. The evidence gathered across all three stages is directionally consistent with this prediction: LM05 exhibited a higher average QBER than BB84 on the noisy simulator (0.0652 versus 0.0650), and a higher QBER on real hardware at the largest and most reliable sample size tested (0.79% versus 0.41% at $N = 500$ on `ibm_fez`). However, a formal two-proportion significance test on this real-hardware result found $p \approx 0.64$, meaning the observed difference does not reach conventional statistical significance at this sample size — a limitation traced directly to LM05’s structurally smaller Control-Mode testing sample relative to BB84’s sifted key, for any fixed number of raw qubits transmitted.

The honest conclusion, stated with the precision this report has aimed to maintain throughout, is therefore that this project’s results are *consistent with, but do not statistically confirm*, the theoretical prediction that LM05 underperforms BB84 under realistic noise. Achieving statistical confirmation would require substantially more raw qubits than the 500 used in this project’s largest run — a specific, quantifiable extension of this work, constrained here by the compute-time limits of the IBM Quantum Open Plan’s free tier. Beyond this central statistical finding, this project additionally confirmed that both protocols were correctly implemented at the circuit level (via an exact, deterministic zero-QBER result in the absence of noise), and it explicitly documented two boundaries of its own scope: that its LM05 implementation represents the protocol’s round trip as a gate-sequence abstraction within a single circuit rather than a genuine physical two-hop channel transmission, and that no active eavesdropping scenario was implemented or evaluated. A report making claims beyond what this evidence supports would not withstand scrutiny; the claims made here are calibrated to what the data, examined rigorously, actually shows

References

- [1] C. H. Bennett and G. Brassard, “Quantum Cryptography: Public Key Distribution and Coin Tossing,” *Proceedings of IEEE International Conference on Computers, Systems and Signal Processing*, Bangalore, India, pp. 175–179, 1984.
- [2] M. Lucamarini and S. Mancini, “Secure Deterministic Communication Without Entanglement,” *Physical Review Letters*, vol. 94, 140501, 2005.
- [3] S. Kumar, S. P. Dash, and G. C. Alexandropoulos, “Performance Analysis of One- and Two-way DV-QKD with MIMO FSO Communication Systems,” presented at *IEEE International Conference on Communications (ICC)*, May 2026. Used strictly for LM05 protocol-logic reference (Control Mode/Message Mode structure, identity/Pauli-Y encoding convention), not cited for any MIMO, FSO channel, or experimental-hardware results.
- [4] W. K. Wootters and W. H. Zurek, “A Single Quantum Cannot Be Cloned,” *Nature*, vol. 299, pp. 802–803, 1982.
- [5] V. Scarani, H. Bechmann-Pasquinucci, N. J. Cerf, M. Dušek, N. Lütkenhaus, and M. Peev, “The Security of Practical Quantum Key Distribution,” *Reviews of Modern Physics*, vol. 81, pp. 1301–1350, Sep. 2009.
- [6] G. Brassard and L. Salvail, “Secret-Key Reconciliation by Public Discussion,” *Advances in Cryptology — EUROCRYPT ’93*, Lecture Notes in Computer Science, vol. 765, pp. 410–423, 1993.
- [7] C. H. Bennett, G. Brassard, C. Crépeau, and U. M. Maurer, “Generalized Privacy Amplification,” *IEEE Transactions on Information Theory*, vol. 41, no. 6, pp. 1915–1923, 1995.
- [8] D. Mayers, “Unconditional Security in Quantum Cryptography,” *Journal of the ACM*, vol. 48, no. 3, pp. 351–406, 2001.
- [9] P. W. Shor and J. Preskill, “Simple Proof of Security of the BB84 Quantum Key Distribution Protocol,” *Physical Review Letters*, vol. 85, no. 2, pp. 441–444, 2000.
- [10] E. B. Wilson, “Probable Inference, the Law of Succession, and Statistical Inference,” *Journal of the American Statistical Association*, vol. 22, no. 158, pp. 209–212, 1927.
- [11] IBM Quantum Platform / Qiskit Runtime documentation, <https://quantum.cloud.ibm.com/docs>, accessed 2026.
- [12] Qiskit Aer documentation, <https://qiskit.github.io/qiskit-aer/>, accessed 2026.

Code Availability

The complete implementation — BB84 and LM05 circuits, Level 1–3 notebooks, and all analysis code — is available at: https://github.com/sajjanmdb123-cmd/Gate-Implementation_BB84_LM05_QKD

How to Run on IBM Quantum Hardware

1. Create a free account at <https://quantum.cloud.ibm.com> and generate an API key from the dashboard.
2. Copy your account's CRN (Cloud Resource Name) from the same dashboard.
3. In the code, save your credentials once:

```
from qiskit_ibm_runtime import QiskitRuntimeService

QiskitRuntimeService.save_account(
    channel="ibm_quantum_platform",
    token="PASTE_YOUR_API_TOKEN_HERE",
    instance="PASTE_YOUR_CRN_HERE",
    overwrite=True
)
```

4. Load the service and select a backend:

```
service = QiskitRuntimeService()
backend = service.least_busy(operational=True, simulator=False)
```

5. Transpile and submit circuits using `SamplerV2`, as detailed in Section 3.2.